



(An Autonomous Institute affiliated to Savitribai Phule Pune University)

CountMinSketch

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Code:

```
#include <iostream> using
namespace std;

class CountMinSketch {
    int **table;
    int d, w;

public:
    // Constructor
    CountMinSketch(int depth, int width) {
        d = depth;    w = width;

        table = new int*[d];
```

```

        for (int i = 0; i < d; i++) {
table[i] = new int[w];          for
(int j = 0; j < w; j++) {
table[i][j] = 0;
        }
    }
}

```

```

// Simple hash functions    int hash(int item, int i) {
return (item + i * 31) % w; // Different hash per row
}

```

```

// Insert element    void
insert(int item) {        for (int i
= 0; i < d; i++) {        int
index = hash(item, i);
table[i][index]++;
    }
}

```

```

// Query frequency
int getCount(int item) {
int min = 999999;

```

```

    for (int i = 0; i < d; i++) {
int index = hash(item, i);

```

```

if (table[i][index] < min) {
    min = table[i][index];
}
}
return min;
}

```

```

// Display table    void display() {
cout << "\nCount-Min Sketch Table:\n";

    for (int i = 0; i < d; i++) {
for (int j = 0; j < w; j++) {
cout << table[i][j] << " ";

        }

        cout << endl;
    }
}

```

```

// Destructor

~CountMinSketch() {

    for (int i = 0; i < d; i++) {
delete[] table[i];

    }

delete[] table;

    }

};

```

```

// Main function

int main() {    int
depth, width;

    cout << "=== Count-Min Sketch Implementation ===" << endl;
cout << "Enter depth (d): ";    cin >> depth;    cout << "Enter
width (w): ";    cin >> width;

    CountMinSketch cms(depth, width);

    int choice;
int item;

    while (true) {        cout << "\n--- Menu ---" << endl;
cout << "1. Insert an element" << endl;        cout << "2.
Query frequency of an element" << endl;        cout << "3.
Display the sketch table" << endl;        cout << "4. Exit"
<< endl;        cout << "Enter your choice: ";        cin >>
choice;

        switch (choice) {
case 1:
            cout << "Enter element to insert: ";
            cin >> item;            cms.insert(item);            cout <<
"Element " << item << " inserted successfully." << endl;
            break;
case 2:

```

```
        cout << "Enter element to query: ";

cin >> item;

        cout << "Estimated count of " << item << ": " << cms.getCount(item) <<
endl;

        break;

case 3:

cms.display();

break;

case 4:

        cout << "Exiting... Thank you!" << endl;

        return 0;


default:

        cout << "Invalid choice. Please try again." << endl;

        }

    }

    return 0;

}
```

Output:

=== Count-Min Sketch Implementation ===

Enter depth (d): 4

Enter width (w): 20

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 1

Enter element to insert: 42

Element 42 inserted successfully.

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 1

Enter element to insert: 42

Element 42 inserted successfully.

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 1

Enter element to insert: 42

Element 42 inserted successfully.

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 2

Enter element to query: 42

Estimated count of 42: 3

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 1

Enter element to insert: 99

Element 99 inserted successfully.

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 1

Enter element to insert: 99

```
Element 99 inserted successfully.
```

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 2

Enter element to query: 99

Estimated count of 99: 2

--- Menu ---

1. Insert an element
2. Query frequency of an element
3. Display the sketch table
4. Exit

Enter your choice: 3

Count-Min Sketch Table:

0 0 3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2

0 0 0 0 0 0 0 0 0 0 2 0 0 3 0 0 0 0 0 0

0 2 0 0 3 0 0 0 0 0 0 0 0 0 0 0 0 0

0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 3 0 0 0 0